

Data Analytics Mini Project Guidelines

Project Title-

Project Objective

The mini project aims to assess your ability to perform real-world data analysis using **Excel** for data cleaning and transformation, and **Power BI** for data visualization and report building.

This mini project carries a total of 25 marks.

Project Scope

Your project will involve the following key components:

1. Data Pre-processing using Excel

- Perform cleaning and transformation tasks on raw data, such as:
 - Removing duplicates
 - Handling missing values
 - Standardizing formats
 - Creating calculated fields (if required)
 - Filtering and sorting
 - Creating Pivot Tables (if required)

2. Data Visualization using Power BI

- Build a report/dashboard with:
 - Multiple visualizations that communicate insights clearly
 - Appropriate use of charts (bar, line, pie, slicers, cards, etc.)
 - Clear and interactive layout (using filters, slicers, drill-downs, etc.)
 - Proper titles, labels, and legends
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Project Domains

Here are some sample domain ideas to help you get started. However, you are free to choose any domain or topic of your interest.

1. Sales & E-commerce

- Sales performance dashboard
- Customer segmentation based on region or product category
- Trend and seasonality analysis of sales data

2. Finance

- Expense tracking dashboard
- Budget vs. Actual analysis
- Profitability by product or region

3. Healthcare

- Hospital/clinic patient records analysis
- Disease count trends across months/years
- Treatment cost analysis

4. Retail

- Inventory and stock analysis
- Customer purchasing patterns
- Store-wise performance comparison

5. Education

- Student performance dashboard
- Course enrollment and completion analysis
- Feedback and satisfaction trends

6. Human Resources (HR)

- Employee attrition or retention reports
- Department-wise performance or salary comparison
- Leave and attendance trends

7. Environmental

- Traffic pattern analysis across regions or time periods

- Air quality monitoring and pollutant trends
- Water consumption or waste generation tracking

8. Social Media & Public Sentiment

- Sentiment analysis of social media posts or movie reviews (e.g., Twitter, IMDB)
- Trending hashtags or topics analysis
- Public opinion analysis on a campaign or event

Dataset Requirements

The dataset should have:

- **Minimum of 500 records**
 - **At least 10 features/columns** (preferably a mix of numerical and categorical data)
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Data Sources

Here are some reliable sources where you can find datasets for your

- project:
1. **UCI Machine Learning Repository:** [UCI Repository](#)
 2. **Google Dataset Search:** [Google Dataset Search](#)
 3. **Data.gov (U.S.):** [Data.gov](#)
 4. **Data.gov.in (India):** [Data.gov.in](#)
 5. **World Health Organization (WHO):** [WHO Data](#)
 6. **India Data Portal:** [India Data Portal](#)
 7. **Awesome Public Datasets:** [Awesome Public Datasets](#)
 8. **OpenML:** [OpenML](#)
 9. **FiveThirtyEight:** [FiveThirtyEight Datasets](#)
 10. **Hugging Face** <https://huggingface.co/datasets>
 11. <https://indiadataportal.com/>
 12. learn.microsoft.com

Submission Requirements

You must submit the following files:

1. **Original Dataset Source**

- Provide the dataset link from where the raw data was sourced.

2. **Cleaned Dataset**

- Submit the **.xlsx** Excel file after performing data cleaning and transformation.

3. **Power BI Report (PDF Export)**

- Export and submit your Power BI report/dashboard as a **PDF file (.pdf)**.

Mini-Project Rubrics [Total: 25 marks]

TASK 1: Dataset Selection (5 marks)

- Relevance, size, and source of the dataset.

TASK 2: Data Pre-processing using Excel (10 marks)

- Cleaning, transformation, and Excel features were used.

TASK 3: Data Visualization using Power BI (10marks)

- Visualizations, insights, and design of the report/dashboard.
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